

Vlad-Emil Flore

Backend Software Engineer (Python, Java, GenAI)

DETAILS

flore.vlad@gmail.com

Barcelona

Romania

EDUCATION

Machine Learning Specialization
DeepLearning.AI & Stanford Online
2026

Master of Science in Informatics
West University of Timișoara
2008 — 2010
Focus on Artificial Intelligence and Distributed Computing
Grade: 9/10

Bachelor of Science in Informatics
West University of Timișoara
2004 — 2008
Grade: 9.67/10

PROFILE

- Backend engineer specializing in **Python and Java**, building cloud-native services and modernizing legacy systems
- Builds **production GenAI systems** for medical coding and document processing, using LLMs, RAG search, and human-in-the-loop review
- Experienced in event-driven microservices on Kubernetes and AWS, technical leadership, and mentoring
- Open to remote and hybrid roles

EMPLOYMENT HISTORY

INDEPENDENT SOFTWARE ENGINEER

Freelancing, Remote (Barcelona)
January 2025 - present

- Developed and maintained **scalable Web APIs** in Python and Java, including integrations with external platforms and services
- Designed and built an **AI medical coding** platform, live with multiple organizations, that turns free-text diagnoses into ICD-9 and ICD-10 codes using **LLM- and RAG-powered** search, with around 85% reported accuracy, higher on ICD-10
- Built the Web API for a live **GenAI document understanding** platform, validating processing pipeline definitions and managing pipeline templates
- Implemented **human-in-the-loop** review, letting experts check, correct, and confirm low-confidence AI results, with corrections feeding back to improve future predictions
- Delivered both platforms as secure, containerized services with pluggable components, automated releases, and thorough documentation

LANGUAGES

Romanian	NATIVE
English	C1
German	B2+
Spanish	A2

SKILLS

- Programming Languages (Python, Java)
- Machine Learning and Generative AI
- Testing (JUnit, Mockito, AssertJ, Pytest, Testcontainers)
- Coding Practices (Clean Code, TDD, SOLID)
- Web API (RESTful APIs, GraphQL, OpenAPI, Swagger, PostGraphile)
- Web Frameworks (Spring Framework, Spring Boot, Django, FastAPI)
- Messaging and Event-Sourcing Systems (RabbitMQ, EventStoreDB)
- Databases & ORM (SQL, PostgreSQL, Oracle, Spring Data JPA, Hibernate, SQLAlchemy)
- Cloud & Infrastructure (AWS, Kubernetes, Keda, Docker, Linux)
- Build Tools (Maven, Gradle, Bazel)

- Worked as a **Java Domain Expert** through Mercor, contributing to AI training, code evaluation, and technical assessment workflows for LLM systems
- Led documentation modernization efforts, including migration to modern documentation platforms to improve maintainability, structure, and multi-version documentation support
- Created software engineering **educational content** on Python, Java, Data Structures & Algorithms, and Git, including animated video series, interactive quizzes, and flashcards

TECH STACK: Python, FastAPI, Django, Java, Spring (Framework, Boot, AI), PyScript, Manim, OpenAPI, Generative AI (LLM, RAG, Agentic Systems), LangChain(4j), LangGraph, LangSmith, Ollama, Weaviate, PostgreSQL (with pgvector), MySQL, Azure Cosmos DB, SQLAlchemy, Alembic, Azure Blob Storage, OAuth, OIDC, JWT, OpenTelemetry, Docker (Compose), Kubernetes, GitHub Actions, AWS (EC2, S3, IAM, Lambda, ECR, ECS, EKS), Antora, Docusaurus

SOFTWARE ENGINEER

Employment, ExB Group, Remote (Nürnberg, Barcelona)

March 2021 — November 2024

- Implemented key features in Python and Java in a team of 4-5, on an event-driven Intelligent Document Processing application of around 15 microservices
- Designed and developed decoupled microservices for document upload, archiving and download, image processing, REST and GraphQL APIs, notifications, and AI model deployment
- Deployed services on Docker and Kubernetes, and diagnosed resource consumption issues in production
- Led internal technical workshops focused on clean code principles and improving overall code quality
- Conducted regular code reviews and participated in pair programming to foster collaboration and knowledge sharing
- Advocated for tooling and workflow improvements to streamline the developer experience and reduce friction in daily tasks

TECH STACK: Python, Java, Django, Spring Framework, Spring Boot, Akka, GraphQL, PostGraphile, PostgreSQL, EventStoreDB, RabbitMQ, Docker, Kubernetes, Helm, Keda, AWS S3, MinIO, AWS ECR, AWS ECS, AWS EKS, Git, GitLab, TeamCity, Argo CD, Grafana, AsciiDoc, Antora, Linux, JUnit

CI/CD (Git, GitHub, GitHub Actions, GitLab, TeamCity, Argo CD, Jenkins, Helm, Ansible)

Team Leadership, Mentoring & Knowledge Sharing

LINKS

Certifications

LinkedIn

GitHub

EDUCATIONAL CONTENT CREATOR

Contract, Hyperskill, Remote (Barcelona)

October 2023 — March 2024

- Created and enhanced educational content for the Spring Boot track offered by JetBrains Academy through the Hyperskill platform, contributing to high-quality, hands-on learning experiences for aspiring developers

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TECH STACK: Java, Spring Framework, Spring Boot

SOFTWARE DEVELOPER / SENIOR TECHNICAL CONSULTANT

Employment, virtual7 GmbH (Nürnberg) and virtual7 Romania (Timișoara)

February 2008 — May 2020

- Consultant on client software projects for public sector and enterprise clients, working in agile teams across the full SDLC

> ### LEAD SOFTWARE DEVELOPER
>
> virtual7 client: IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)
>
> *September 2019 — April 2020*
>
> - Served as Lead Developer for one of two agile development teams, guiding technical direction and team collaboration
> - Defined and evolved the application's architecture to support scalability and maintainability
> - Promoted clean code and consistent development practices to maintain code quality
> - Collaboratively decomposed features into manageable tasks, ensuring effective sprint planning and delivery
> - Implemented and tested application features across the full stack, maintaining close alignment with business requirements
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> **TECH STACK:** Java SE 8, Java EE 7, Angular, TypeScript, JavaScript, JUnit, Cucumber, Jenkins, Docker, Linux, Oracle Database, Git
>
> ### FULL STACK DEVELOPER
>
> virtual7 client: Bundesamt für Migration und Flüchtlinge (BAMF), on-site (Nürnberg)
>
> *May 2018 — September 2019*
>
> - Implemented features across both frontend and backend of a web application

- Improved and maintained project documentation to support developer onboarding and knowledge sharing
- Managed DevOps-related tasks and supported deployment activities to ensure smooth delivery and system stability

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TECH STACK: Java SE 8, Spring Framework, Spring Boot, JUnit, Angular, TypeScript, JavaScript, Ansible, Jenkins, Docker, Oracle Database, Hibernate, Linux, Git

SOFTWARE DEVELOPER

virtual7 client: IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)

May 2014 — February 2018

- Developed a publishing workflow for web content within Oracle CMS, streamlining the content creation and approval process
- Extended the CMS by implementing custom tools and features to support efficient content management and improve the user experience for content editors

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TECH STACK: Oracle WebCenter Content, Java, HTML, CSS, JavaScript, Oracle Database, Linux, CVS, Git